OPLOGS-06: Topology & Entity Context

> **Series:** OPLOGS — OpenPipeline Logs | **Notebook:** 6 of 8 | **Created:** December 2025 | **Last Updated:** 10/12/2026

Leveraging Entity Relationships in Log Analysis

This notebook explores how Dynatrace enriches logs with entity context (hosts, processes, services, Kubernetes) for topology-aware analysis.

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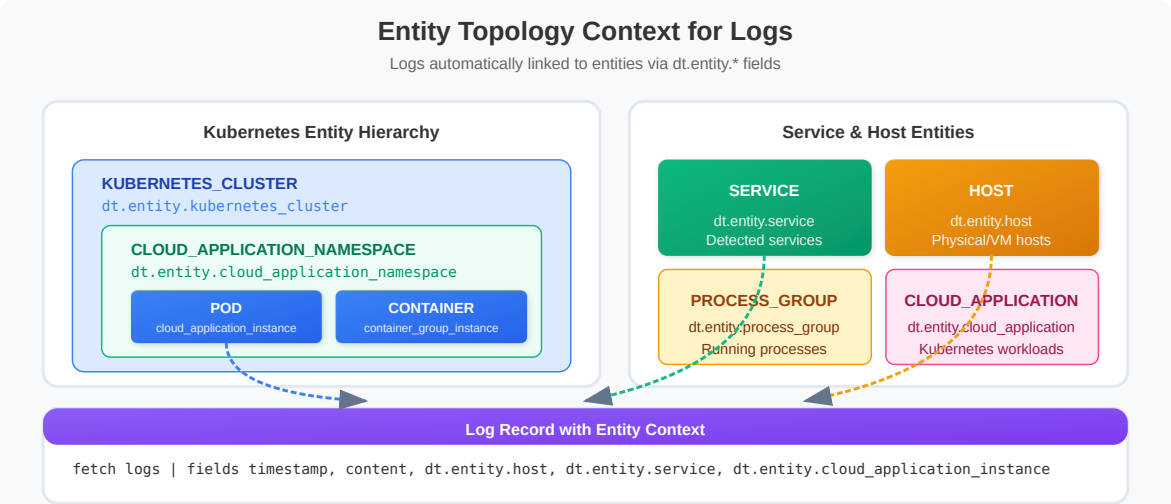
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Prerequisites

- ✅ Access to a Dynatrace environment with log data
- ✅ Completed OPLOGS-01 through OPLOGS-05
- ✅ Understanding of Dynatrace entity model (helpful)

1. Entity Types Overview

Dynatrace automatically enriches logs with entity context:



Entity Field	Description	Example
`dt.entity.host`	Host entity ID	`HOST-ABC123`
`dt.entity.process_group`	Process group ID	`PROCESS_GROUP-XYZ789`
`dt.entity.process_group_instance`	PGI ID	`PROCESS_GROUP_INSTANCE-DEF456`
`dt.entity.service`	Service entity ID	`SERVICE-QRS012`
`dt.entity.kubernetes_cluster`	K8s cluster ID	`KUBERNETES_CLUSTER-TUV345`

> **Classic entity fields are deprecated; Smartscape fields are the stable successors.** In the semantic dictionary, `dt.entity.host`, `dt.entity.service`,

> `dt.entity.process_group`, `dt.entity.process_group_instance` and `dt.entity.kubernetes_cluster` are all marked `deprecated`. Their Smartscape counterparts — `dt.smartscape.host`, `dt.smartscape.service`, `dt.smartscape.process`, `dt.smartscape.k8s_cluster`, `dt.smartscape.k8s_pod` — are `stable` and already populated on logs. The queries below still use the classic fields because they keep working, but coverage of the two families is not identical (on the validation tenant over 15 minutes: 166k records with `dt.entity.host`, 156k with `dt.smartscape.host`). Run the coverage query below before you move a dashboard or alert to the Smartscape fields, and see FAQ-16 for the classic-to-Smartscape mapping.
>
> **Dictionary:** `dt.entity.host` / `dt.entity.service` / `dt.entity.process_group` / `dt.entity.process_group_instance` / `dt.entity.kubernetes_cluster` (`deprecated`); `dt.smartscape.host` / `dt.smartscape.service` / `dt.smartscape.process` / `dt.smartscape.k8s_cluster` / `dt.smartscape.k8s_pod` (`stable`), read 10/02/2026.

Kubernetes Context Fields

Field	Description
`k8s.namespace.name`	Kubernetes namespace
`k8s.pod.name`	Pod name
`k8s.pod.uid`	Pod unique identifier
`k8s.container.name`	Container name
`k8s.cluster.name`	Cluster name
`k8s.deployment.name`	Deployment name
`k8s.workload.name`	Workload name
`k8s.workload.kind`	Workload type (Deployment, StatefulSet, etc.)

```
// Discover available entity fields in your logs — classic (deprecated) and
Smartscape (stable)
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    with_host = countIf(isNotNull(dt.entity.host)),
    with_smartscape_host = countIf(isNotNull(dt.smartscape.host)),
    with_process_group = countIf(isNotNull(dt.entity.process_group)),
    with_smartscape_process = countIf(isNotNull(dt.smartscape.process)),
    with_service = countIf(isNotNull(dt.entity.service)),
    with_smartscape_service = countIf(isNotNull(dt.smartscape.service)),
    with_k8s_cluster = countIf(isNotNull(dt.entity.kubernetes_cluster)),
    with_k8s_namespace = countIf(isNotNull(k8s.namespace.name))
  }
```

2. Host Topology

Analyze logs by host to understand infrastructure patterns.

```
// Log volume by host
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.host)
| summarize {log_count = count()}, by: {dt.entity.host}
| sort log_count desc
| limit 15
```

```
// Error distribution by host
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.host)
| summarize {
    total = count(),
    errors = countIf(status == "ERROR")   // all error-class levels, incl.
CRITICAL and FATAL
  }, by: {dt.entity.host}
| fieldsAdd error_rate = (errors * 100.0) / total
| sort errors desc
| limit 15
```

```
// Host with log.source breakdown
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.host)
| summarize {count = count()}, by: {dt.entity.host, log.source}
| sort count desc
| limit 20
```

3. Process Group Topology

Process groups represent logical application components across hosts.

> **`dt.entity.process_group` is not always a single value.** On the validation tenant (10/02/2026) some log records carried an **array** of process-group IDs, so `by: {dt.entity.process_group}` produces some groups keyed by a whole array, and `== "PROCESS_GROUP-…"` misses those records. Where that matters, `expand dt.entity.process_group` before grouping, or filter with `in("PROCESS_GROUP-…", dt.entity.process_group)`.

```
// Logs by process group
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.process_group)
| summarize {log_count = count()}, by: {dt.entity.process_group}
| sort log_count desc
| limit 15
```

```
// Process group error analysis
// status == "ERROR" covers every error-class level (ERROR, SEVERE, CRITICAL,
FATAL, …);
// loglevel == "ERROR" alone misses SEVERE and the rest
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.process_group)
| filter status == "ERROR"
// Group on a normalized signature (UUIDs, then digits collapsed) so one
recurring error is one row, not one row per second
| fieldsAdd error_sig = substring(replacePattern(replacePattern(content, "[0-
9a-fA-F]{8} '-' [0-9a-fA-F]{4} '-' [0-9a-fA-F]{4} '-' [0-9a-fA-F]{4} '-' [0-
9a-fA-F]{12}", replacement: "&lt;id&gt;"), "[0-9]+", replacement: "#"), from:
0, to: 80)
| summarize {error_count = count(), sample = takeAny(substring(content, from:
0, to: 100))}, by: {dt.entity.process_group, error_sig}
| sort error_count desc
| limit 20
```

```
// Process group to host mapping
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.process_group) AND isNotNull(dt.entity.host)
| summarize {count = count()}, by: {dt.entity.process_group, dt.entity.host}
| sort count desc
| limit 20
```

4. Kubernetes Topology

OpenPipeline enriches container logs with rich Kubernetes context.

```
// Logs by Kubernetes namespace
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {log_count = count()}, by: {k8s.namespace.name}
| sort log_count desc
| limit 15
```

```
// Kubernetes namespace with error breakdown
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {
    total = count(),
    errors = countIf(status == "ERROR"),
    warnings = countIf(status == "WARN")
  }, by: {k8s.namespace.name}
| fieldsAdd error_percentage = round((errors * 100.0) / total, decimals: 2)
| sort errors desc
| limit 10
```

```
// Pod-level analysis
fetch logs, from: now() - 1h
| filter isNotNull(k8s.pod.name)
| summarize {
    log_count = count(),
    error_count = countIf(status == "ERROR")
  }, by: {k8s.namespace.name, k8s.pod.name}
| sort error_count desc
| limit 20
```

```
// Workload analysis (Deployments, StatefulSets, etc.)
fetch logs, from: now() - 1h
| filter isNotNull(k8s.workload.name)
| summarize {
    log_count = count(),
    unique_pods = countDistinct(k8s.pod.name)
  }, by: {k8s.namespace.name, k8s.workload.kind, k8s.workload.name}
| sort log_count desc
| limit 15
```

```
// Container-level detail
fetch logs, from: now() - 1h
| filter isNotNull(k8s.container.name)
| summarize {log_count = count()}, by: {k8s.namespace.name, k8s.pod.name,
k8s.container.name}
| sort log_count desc
| limit 20
```

5. Service Mapping

Connect logs to Dynatrace-detected services for full observability.

```
// Logs by service entity
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.service)
| summarize {log_count = count()}, by: {dt.entity.service}
| sort log_count desc
| limit 15
```

```
// Service error rates from logs
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.service)
| summarize {
    total = count(),
    errors = countIf(status == "ERROR")
  }, by: {dt.entity.service}
| fieldsAdd error_rate = round((errors * 100.0) / total, decimals: 2)
| sort error_rate desc
| limit 15
```

```
// Service to process group relationship
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.service) AND isNotNull(dt.entity.process_group)
| summarize {count = count()}, by: {dt.entity.service,
dt.entity.process_group}
| sort count desc
| limit 20
```

6. Cross-Entity Correlation

Use entity context to correlate logs across the topology.

```
// Full topology view: Cluster &gt; Namespace &gt; Pod &gt; Container
fetch logs, from: now() - 1h
| filter isNotNull(k8s.cluster.name)
| summarize {log_count = count()}, by: {
    k8s.cluster.name,
    k8s.namespace.name,
    k8s.workload.name,
    k8s.pod.name
  }
| sort log_count desc
| limit 25
```

```
// Entity coverage report
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    host_coverage = round((countIf(isNotNull(dt.entity.host)) * 100.0) /
count(), decimals: 1),
    pg_coverage = round((countIf(isNotNull(dt.entity.process_group)) * 100.0)
/ count(), decimals: 1),
    service_coverage = round((countIf(isNotNull(dt.entity.service)) * 100.0) /
count(), decimals: 1),
    k8s_coverage = round((countIf(isNotNull(k8s.namespace.name)) * 100.0) /
count(), decimals: 1)
  }
```

```
// Logs without entity context (potential configuration issue)
fetch logs, from: now() - 1h
| filter isNull(dt.entity.host) AND isNull(dt.entity.process_group)
| summarize {orphan_count = count()}, by: {dt.openpipeline.source}
| sort orphan_count desc
```

```
// Trace correlation: Logs with trace context
fetch logs, from: now() - 1h
| filter isNotNull(trace_id) OR isNotNull(span_id)
| summarize {
    logs_with_trace = count(),
    unique_traces = countDistinct(trace_id)
  }, by: {k8s.namespace.name}
| sort logs_with_trace desc
| limit 10
```

7. Using Entity IDs for Lookups

Entity IDs enable cross-data-type correlation.

```
// Get distinct entity IDs per namespace (pick the namespace you care about
from the result)
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {
    unique_hosts = collectDistinct(dt.entity.host),
    unique_pgs = collectDistinct(dt.entity.process_group),
    unique_services = collectDistinct(dt.entity.service)
  }, by: {k8s.namespace.name}
| limit 10
```

```
// Find logs for a specific entity (replace with actual entity ID)
// fetch logs, from: now() - 1h
// | filter dt.entity.host == "HOST-XXXXXX"
// | summarize {count = count()}, by: {loglevel}

// Discovery query to find entity IDs
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.host)
| summarize {sample = takeFirst(dt.entity.host)}, by: {k8s.namespace.name}
| limit 10
```

8. Topology-Based Alerting Patterns

Use entity context to create meaningful alert conditions.

```
// Alert pattern: Errors per namespace (for threshold alerting)
fetch logs, from: now() - 15m
| filter status == "ERROR"
| summarize {error_count = count()}, by: {k8s.namespace.name}
| filter error_count &gt; 10
| sort error_count desc
```

```
// Alert pattern: Hosts with high error rate
fetch logs, from: now() - 15m
| filter isNotNull(dt.entity.host)
| summarize {
    total = count(),
    errors = countIf(status == "ERROR")
  }, by: {dt.entity.host}
| filter total &gt; 100  // Minimum sample size
| fieldsAdd error_rate = (errors * 100.0) / total
| filter error_rate &gt; 5  // Alert if &gt;5% errors
| sort error_rate desc
```

```
// Alert pattern: pod restarts — read the restart metric, not log text
// Counting "started" / "initializing" lines measures how chatty a pod's
startup is, not
// how often it restarted. dt.kubernetes.container.restarts is a delta count,
so summing
// it over the window gives restarts per pod.
timeseries restarts = sum(dt.kubernetes.container.restarts, default: 0), from:
now() - 1h,
  by: {k8s.namespace.name, k8s.pod.name}
| fieldsAdd restart_count = arraySum(restarts)
| filter restart_count &gt; 3   // more than 3 restarts in 1h = potential
crash loop
| fields k8s.namespace.name, k8s.pod.name, restart_count
| sort restart_count desc
```

📝 Summary

In this notebook, you learned:

✅ **Entity types** - HOST, PROCESS_GROUP, SERVICE, KUBERNETES_CLUSTER
✅ **Host topology** - Log volume and errors by host
✅ **Process groups** - Application component analysis
✅ **Kubernetes context** - Namespace, pod, workload, container
✅ **Service mapping** - Connecting logs to detected services
✅ **Cross-entity correlation** - Full topology views
✅ **Alerting patterns** - Threshold-based topology alerts

➡️ Next Steps

Continue to **OPLOGS-07: Analytics & Dashboards** for aggregation and visualization patterns.

📚 References

- Smartscape on Grail (DT docs)
- Set up Dynatrace on Kubernetes (DT docs)
- Connect log data to traces (DT docs)

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